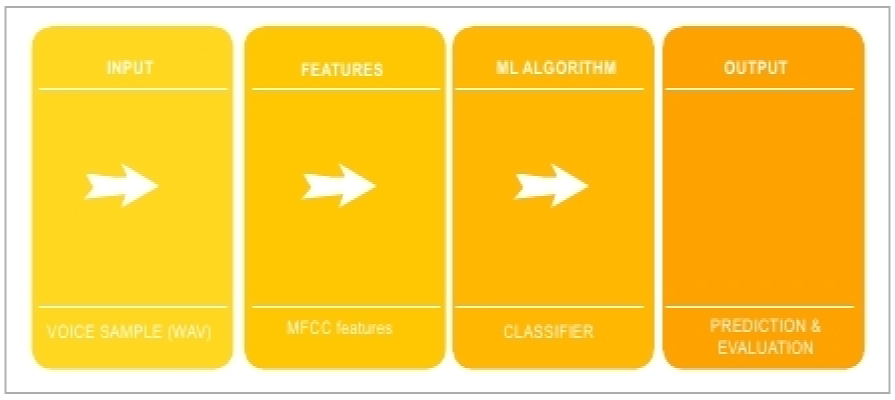

Figure 4: A typical machine learning pipeline

For this task, a voice sample is represented as features. Usually audio data is described using very well known mel-frequency cepstral coefficient (MFCC) features. These coefficients are real numbers which are passed to a classifier as input. The classifier builds a model using this data and uses the model to predict the class of an unknown sample.

Tools for building machine learning applications

This is the era of ML, so plenty of tools are available for it and are also being developed frequently. In Table 1, some tools and software packages prominent in this field have been summarised. Apart from these, there are many dedicated options available for certain domains.

Tool	Functionality/Remarks
Weka	GUI based interface with implemented algorithms
R	A rich package with inbuilt algorithms
Python	A computational package with many rich libraries
Orange	GUI interface (similar to Weka)
RapidMiner	GUI interface (similar to Weka)
MATLAB	The richest computational package, but commercial
Octave	An open source alternative to MATLAB

Table 1: List of machine learning tools

The big question: Can machines outperform humans?

The Hollywood movie, *2001: A Space Odyssey*, which was released in the 1960s, predicted that there would be a time when machines would not only outperform humans but also take charge and start treating humans as their slaves. In the movie, there is a scene in which the machine refuses to execute commands given by a human operator, and instead operates by itself. Similar concepts are also seen in movies like *Wall E* and the *Terminator* series. Recently, Facebook also observed a self-programming phenomenon and suspended its further use. As of now, many tasks cannot be accomplished properly without human intervention. However, the days are not far when machines will be self-sustaining.

Potential applications of machine learning

Machine learning is a multi-disciplinary field, which encompasses several domains including cloud computing, artificial intelligence, probability and statistics, high performance computing, Big Data, genetic algorithms, etc.

The following are some typical applications of ML:

- Speech processing - A speech signal can be analysed to recognise people, messages, gender, emotions, language, etc.
- Music information retrieval - A song can be processed to identify instruments, singers, composers, the *raag*, genre, etc.
- Computer vision - A machine can be used to recognise different objects that are part of the queried image. It can also be used for image tagging automatically. Further, it can be used to predict the gender, age and mood of the person. Pictures similar to the query image can also be obtained.
- Video processing - Analysis of a video may lead to successfully tracking some suspicious event that occurred during a certain time frame, or to detect the presence of a specific person in a video clip.
- Sentiment analysis - Movie or product reviews by users can be analysed to find out their sentiments, which can be used to recommend the product or movie to other users with similar interests. It is known that politicians also have started using the results of such analysis to track the mood of their voters and change the strategy for their election campaign.
- Recommender systems - Systems can understand the preferences of the customers by observing browsing patterns or purchase history and can recommend relevant items to simplify the searching process.
- Anomaly detection - Fraudulent transactions can be revealed with the help of such outlier analysis.
- Document classification - Several Web documents can be classified based on certain terms (keywords).
- Cognitive radio - Intelligent spectrum sensing can be achieved by identifying idle spectrum bands for the optimal use of the spectrum.
- Robotics - Robots can be trained to navigate accurately and then used in space or to perform surgeries.

Many giant companies have set up research labs for machine

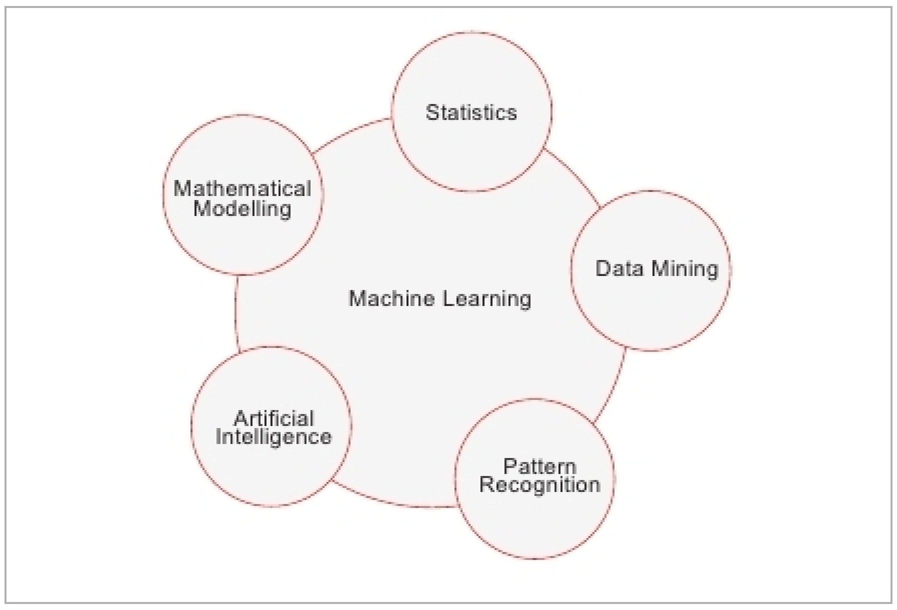

Figure 5: ML, a multi-disciplinary field